SeedingMinds

Assessment Name

Java_S2_030126

Module: Coding

Question 1:

Marks: +10 -0

Early Exit Optimization

In a world where data drives decisions, an e-commerce company constantly seeks ways to optimize its processes to enhance customer experience. One critical aspect of their operations is sorting product listings based on various criteria, such as price, rating or popularity. The existing sorting algorithm is bubble sort, which, while easy to implement, struggles with efficiency, especially when handling large datasets during peak shopping seasons.To address this issue, the company decides to implement an optimization strategy for bubble sort that reduces unnecessary iterations. The optimization is based on the observation that bubble sort can exit early if no swaps occur during a complete pass through the array. This indicates that the array is already sorted, allowing the algorith to terminate early instead of continuing to iterate through the remaining elements.

Constraints:

- The array can contain at most 10^3 integers.
- Each integer in the array can be in the range of 0≤price≤10^6

Example:

Example 1:

Input: 5
5 2 9 1 5
Output: 1 2 5 5 9
4

Example 2:

Input: 3
3 4 5
Output: 3 4 5
1

Explanation:

Your task is to implement bubbleSort() function with early exit optimization in Java. The goal is to ensure that the algorithm exits as soon as the array is sorted, avoiding unnecessary iterations. You will need to track and display the number of iterations taken until th array becomes sorted to demonstrate the effectiveness of the optimization.

Public Test Cases:

#	INPUT	EXPECTED OUTPUT
1	7 11 4 7 3 1 9 6	1 3 4 6 7 9 11 5
2	4 7 6 5 4	4 5 6 7 4

Question 2:

Marks: +10 -0

Q2 - Print all Even Numbers From an Array

Write a Program to Print the all Even Elements from The Given Array?

Constraints:

Input :- First Line of Input Consists of One Integer Value (Array Size).

Second Line of Input Consists of few Integer Values Separated by Spaces (Array Elements).

Output :- Print All the Even Array Elements.

Constraints :- If there is no Even Elements in a Given Array then Print "**No Even Numbers.**".

Example:

Input 1 : 8

86 210 33 73 53 93 848 48

Output 1: 86 210 848 48

Input 2 : 6

1 2 3 4 5 6

Output 2: 2 4 6

Input 3 : 5

5 10 88 2 5

Output 3: 10 88 2

Explanation:

NA

Public Test Cases:

#	INPUT	EXPECTED OUTPUT
1	8 86 210 33 73 53 93 848 48	86 210 848 48
2	6 1 2 3 4 5 6	2 4 6
3	5 5 10 88 2 5	10 88 2

Question 3:

Marks: +10 -0

Q3 - Sum of All Odd Numbers From an Array

Write a Program to Find the sum of all odd numbers from a Given Array?

Constraints:

Input :- First Line of Input Consists of One Integer Value (Array Size).

Second Line of Input Consists of few Integer Values Separated By Spaces (Array Elements).

Output :- Print the Sum of All the Odd Array Elements.

Constraints :- Array Size must be Greater then or equal to 5 or else Print "**InValid Array Size.**".

If there is no Odd Elements in a Given Array then Print "**No Odd Numbers.**".

Example:

Input 1 : 8

8 6 3 53 93 84 210 294

Output 1: 149

Input 2 : 3

1 2 3

Output 2: InValid Array Size.

Explanation:

NA

Public Test Cases:

#	INPUT	EXPECTED OUTPUT
1	8 8 6 3 53 93 84 210 294	149
2	3 1 2 3	InValid Array Size.

Question 4:

Marks: +10 -0

Q3 - Rhombus Pattern

Write a Program to Print the Following Pattern?

If Input is 5 then Print

```
    1
   2 2
  3 3 3
 4 4 4 4
5 5 5 5 5
 4 4 4 4
  3 3 3
   2 2
    1
```

Constraints:

Input :- First Line of Input Consists of One Integer Value.

Output :- Print the Given Pattern.

Constraints :- Given Input must be Greater than 0 or Else Print **"Invalid Input"**.

Example:

Input 1: 5

Output 1:

```
    1
   2 2
  3 3 3
 4 4 4 4
5 5 5 5 5
 4 4 4 4
  3 3 3
   2 2
    1
```

Input 2 : -5

Output 2: Invalid Input

Explanation:

NA

Public Test Cases:

#	INPUT	EXPECTED OUTPUT
1	5	1 2 2 3 3 3 4 4 4 4 5 5 5 5 5 4 4 4 4 3 3 3 2 2 1
2	-5	Invalid Input